

Abstract for the Tsukuba Science Edge 2026 [English form]

Your research stage is : ☐Planning ☐Preliminary ☐Result 【Required: Pick only one】

Area Check Most appropriate one 【Required】

☐Physics ☐Chemistry ☐Biology & Medical ☐Earth Sciences ☐Mathematics & Information

Participant's Information 【Published as are written here in English】

【School / Organization Name】

【Representative's Name (top author)】

【Member's Name】

Supervising Teacher(s) / Person(s) 【Responsible for content guarantee and inquiry response】

【Name】

Abstract of the Presentation

【Title】

【Outline and its First Inspiration】

During on-campus observations, we discovered wild *Boiga kraepelini* and became interested in this species. At the same time, as human activities increasingly damage animals' natural habitats, it is crucial for us to better understand how to coexist with wildlife. We were curious about the differences in locomotor abilities between arboreal snake species and the more commonly observed terrestrial snakes, and thus conducted this experiment.

The experiment consisted of three parts. First, we measured the body dimensions of the experimental snakes, including the lengths and widths of different body segments, and compared the anatomical differences among species. Next, we placed the snakes on a classroom desk and encouraged them to extend the anterior part of their bodies toward another desk, allowing their bodies to hang freely in midair. We then measured the maximum distance they were able to reach between the two desks. Finally, we prepared plastic containers of varying heights to test the snakes' success rates and the time required for upward climbing.

【Novelty of Your Goal and its Original Value】

Existing literature has largely focused on describing snake locomotion patterns, such as serpentine crawling and sidewinding, while only a few studies have conducted quantitative measurements of the limits of vertical lifting and horizontal reaching across multiple snake species together.

In this study, we measured the maximum horizontal reaching distance and vertical extension ability of different snake species to analyze their reaching and spanning performance. The experimental results were then organized and compared with body size to investigate whether body size is correlated with reaching and extension capabilities. In addition, we further compared snakes of different habitats and body sizes to examine differences in their reaching and spanning abilities, and analyzed how these differences may be related to their living environments and behavior.

【Creativity of Your Research Plan】

Rather than relying only on observation, we designed the experiment with height differentials to bring out the snakes' locomotor limits. The experiment consisted of a "vertical extension test" and a "horizontal reaching test," combined with video analysis to observe muscle control and failure patterns (e.g., pulling back due to insufficient tension) during extreme extension.

At the beginning of the experiment, we initially used branches to encourage the snakes to stretch upward. Later, we decided to use containers of varying heights to test whether the snakes could climb out. This adjustment not only increased the convenience of the experiment but also reduced human interference, allowing for more reliable conclusions.

Additionally, using containers provided a clearer demonstration of how humans can safely house and release snakes back into the wild, aligning with the United Nations Sustainable Development Goal 15 (SDG15).

During the experiment, we continuously monitored the snakes' conditions to ensure they were motivated and willing to participate. At the same time, we maintained a safe distance between the snakes to avoid excessive interference.

Submission Deadline : January 12, 2026

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【Feasibility and its Prospect, based on Your Result and Interpretation】

【Future Potential and Your Plan, if any】

Through this experiment, we aim to help people remain calm when facing snakes indoors and to understand their movement range, allowing for proper handling and release. By understanding the snakes' maximum reaching and spanning capabilities, we can design more scientifically informed habitat corridors or snake management structures. For example, if a species' maximum reaching extension ability is 30 cm, designing a 40cm high fence in a community can achieve physical separation without harming the snakes, promoting human–snake coexistence. This project directly addresses SDG 15 (Life on Land) by reducing human-caused mortality and injury in natural ecosystems.

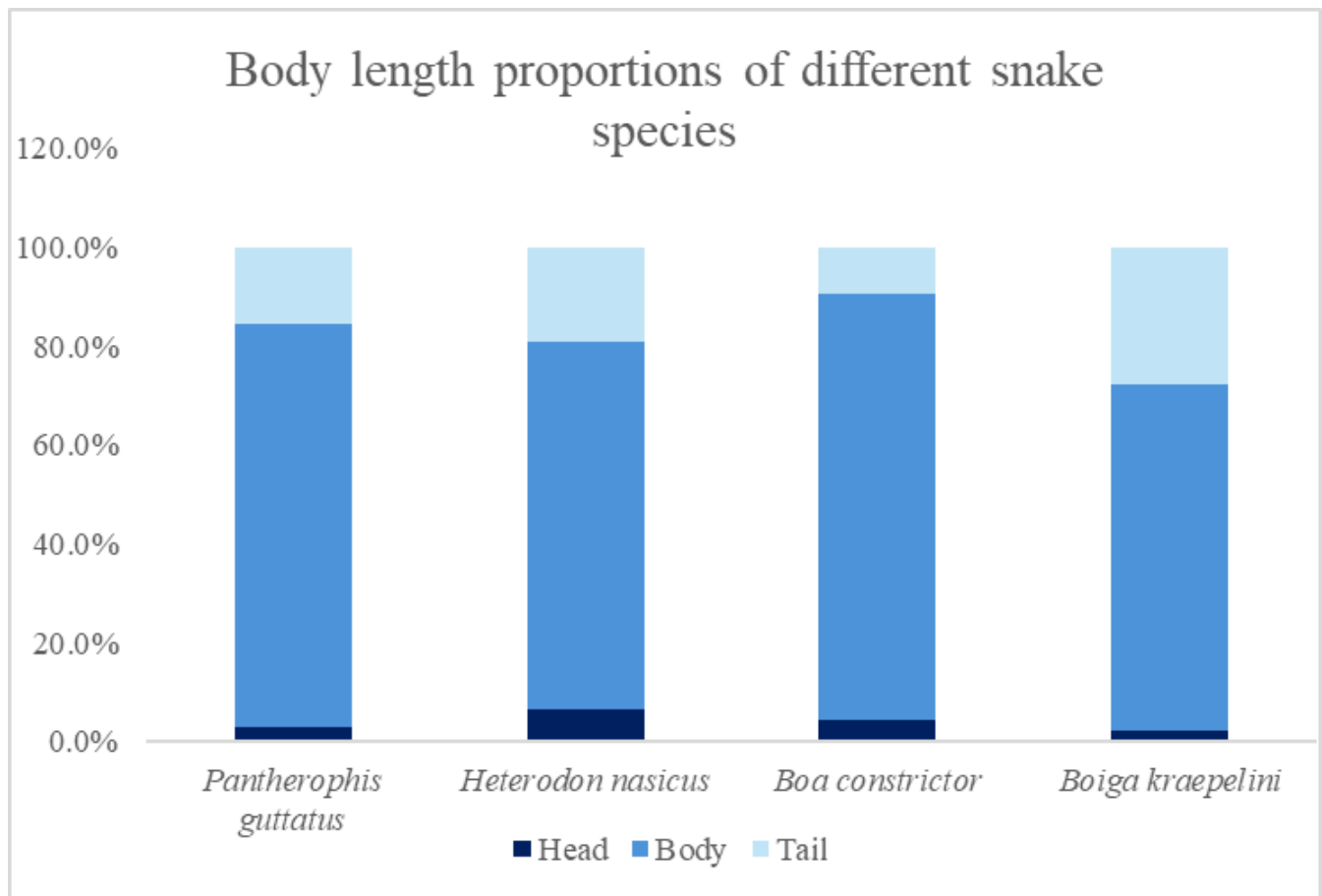
The mechanism by which arboreal snakes anchor their tails to perform cantilevered extension can also inspire the development of rescue robots. By mimicking their center of mass and muscle contraction patterns, robots could navigate discontinuous terrains, such as earthquakes, more effectively for crossing gaps and performing rescue operations.

In the future, we hope to include a greater diversity of snake species to conduct more accurate experiments. We also aim to study how scale structure and body muscle contraction influence locomotor performance and collaborate with experts to develop devices that may lead to safer human–snake coexistence.

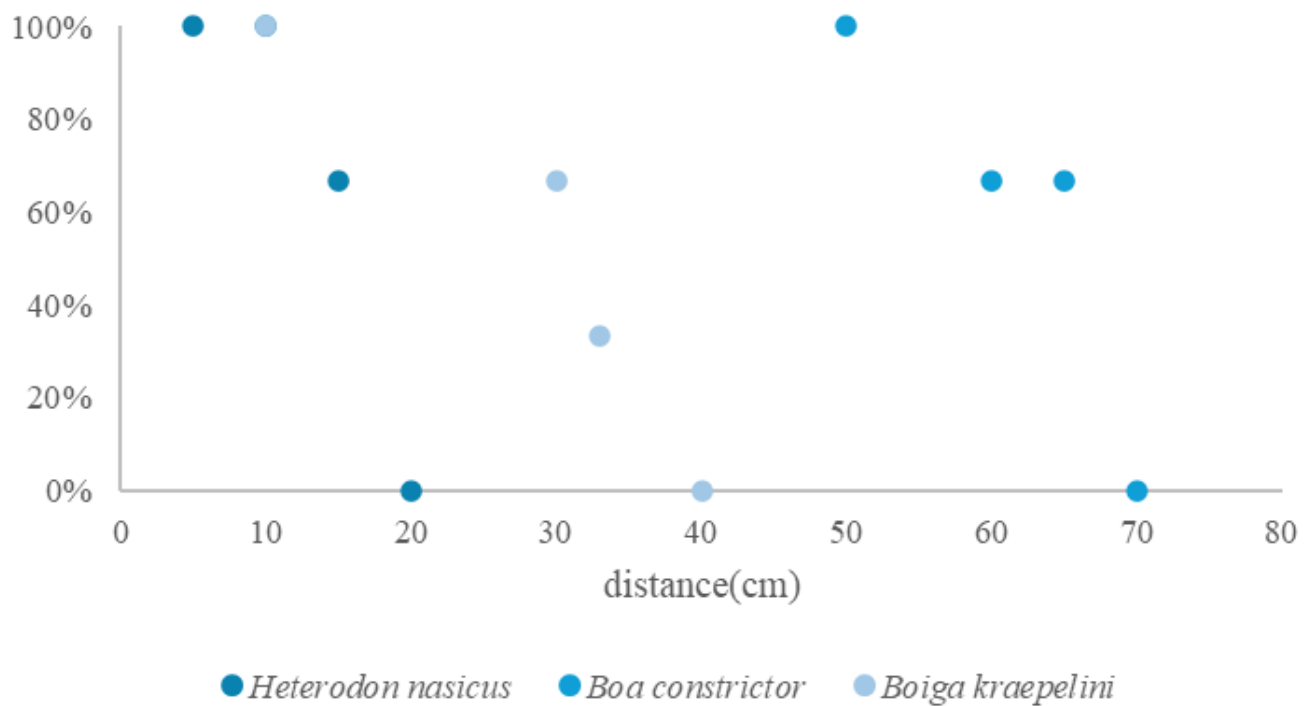
【References】

- 五、Sheehy, Coleman M., Albert, James S., Lillywhite, Harvey B., Van Damme, Raoul (2016). The evolution of tail length in snakes associated with different gravitational environments.
- 六、Stanisław Bury, Bartosz Borczyk, Tomasz Skawiński (2019). Ventral scale width in snakes depends on habitat but not hunting strategy.
- 七、Bruce C Jayne(2020). What Defines Different Modes of Snake Locomotion?
- 八、Julie A Savidge, Thomas F Seibert, Martin Kastner, Bruce C Jayne (2021). Lasso locomotion expands the climbing repertoire of snakes.
- 九、Greg Byrnes, Bruce C Jayne(2012). The effects of three-dimensional gap orientation on bridging performance and behavior of brown tree snakes (*Boiga irregularis*).
- 十、Michelle Graham, John J Socha(2021). Dynamic movements facilitate extreme gap crossing in flying snakes.
- 十一、Bruce C Jayne, Steven J Newman, Michele M Zentkovich, H Matthew Berns(2015). Why arboreal snakes should not be cylindrical: body shape, incline and surface roughness have interactive effects on locomotion.
- 十二、李家萱、蔣明蕙 (2001) , 蛇類體長大小與跨越能力關係之探討。臺北市立第一女子高級中學。

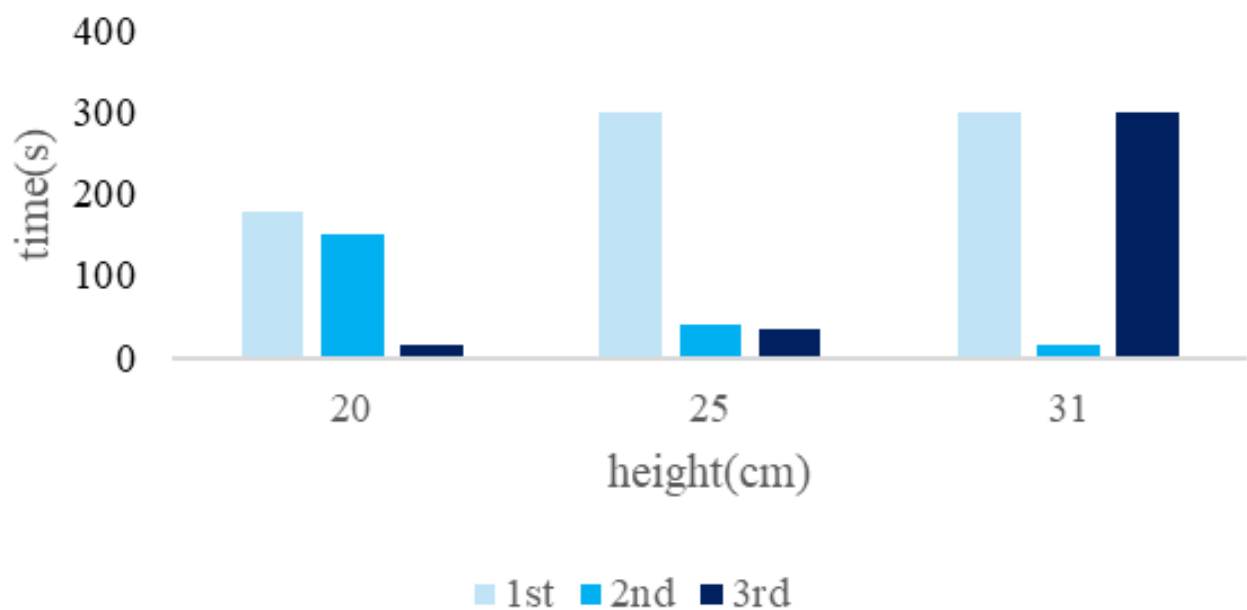
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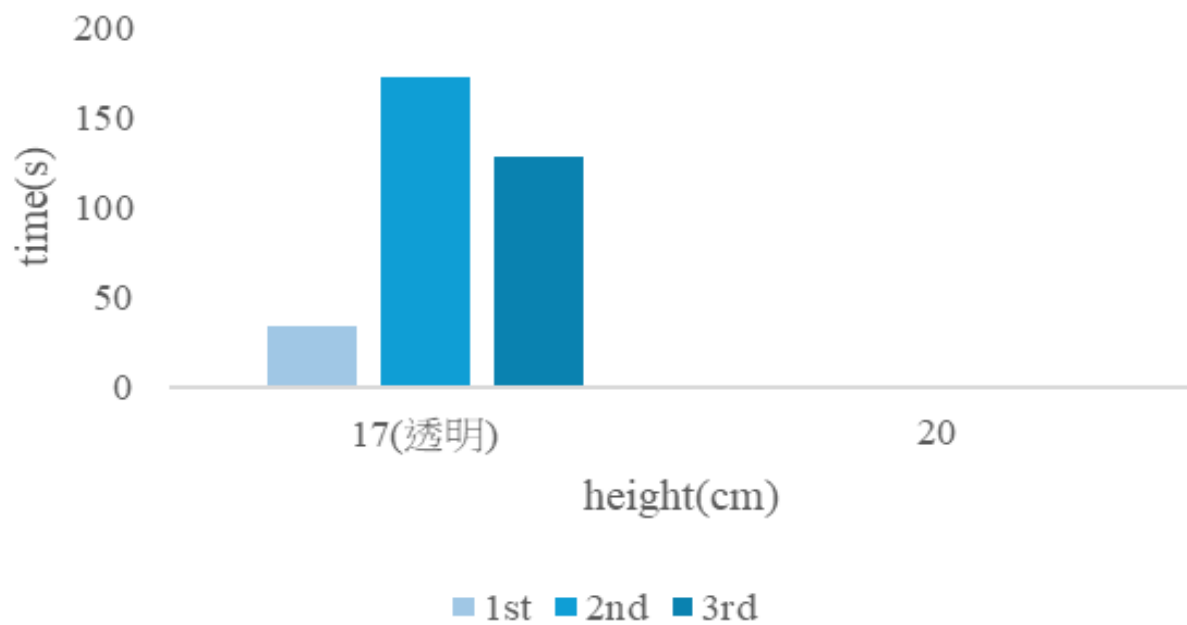
Horizontal reaching success rate of different snakes chart



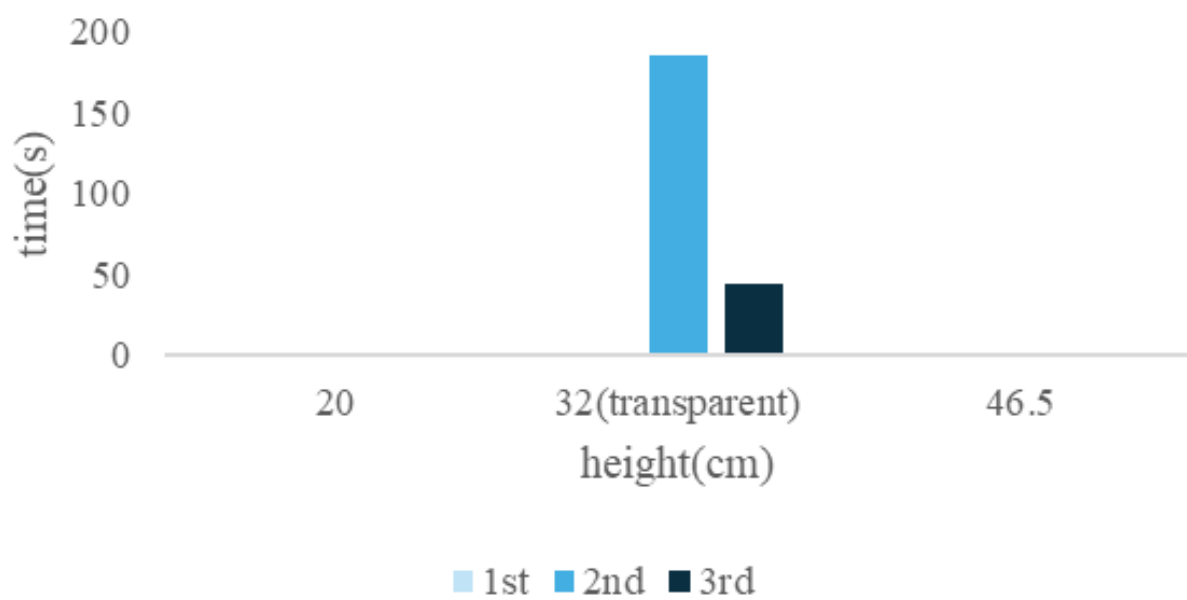
Pantherophis guttatus vertical extensio accomplish time chart



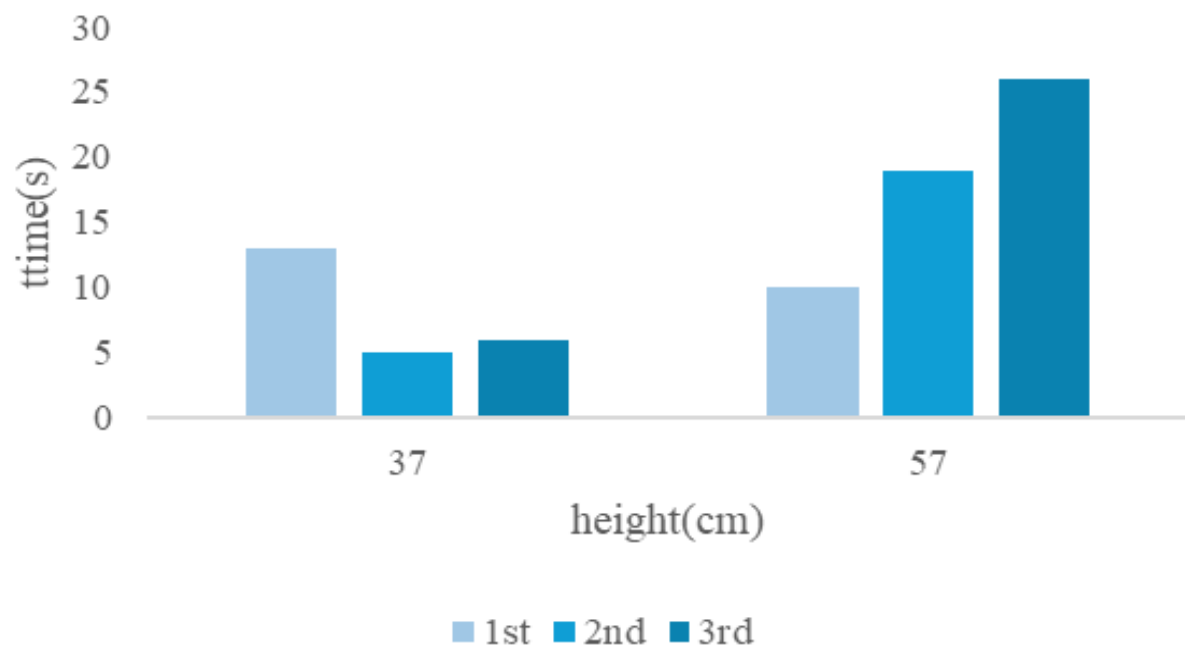
Heterodon nasicus vertical extensio
accomplish time chart



Boa constrictor vertical extensio accomplish
time chart



Boa constrictor vertical extensio accomplish
time chart (box)



The proportion of body length in horizontal reach
and vertical extension among different snakes

